Wavvar MMWK Module Documentation

PRO Module Introduction

Wavvar Technologies

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1. Module Overview

PRO (RTP) is part of the RPX 6843 sensing module family. The 6843 series is Wavvar's flagship module product line, built on TI's high-performance millimeter-wave radar technology for advanced spatial sensing scenarios that require stable motion tracking and accurate spatial measurement.

Compared with MINI, PRO uses an ESP32-S3 as the main controller; all other functional capabilities remain aligned with MINI.

Typical applications include:

- Human presence detection and fall detection
- Dynamic trajectory tracking
- Occupancy detection
- Activity recognition
- Entry and exit monitoring
- In-bed / out-of-bed detection
- Point cloud data visualization

2. Appearance and Dimensions

PRO module appearance reference

Item	Specification
Dimensions	83x83x17 mm

3. Technical Specifications and Key Features

Category	Item	Specification
Power	External Supply	5V DC 2A
	Adapter	100-240V AC input
	System Power Consumption	< 10W
Operating Parameters	Installation Method	Ceiling mount or wall mount
	Maximum Detection Range (Wall-mounted)	Depends on installation height, tilt angle, target reflection characteristics, and algorithm configuration; configurable
	Field of View (FOV)	Approximately 120°-140° horizontally (estimated from the antenna pattern; actual coverage depends on installation method, enclosure structure, and algorithm configuration)
	Operating Temperature	0°C to 45°C (system ambient temperature)
	Operating Humidity	< 95% (non-condensing)
	Wall-mount Pitch Angle	30° (downward tilt)
Radar Characteristics	RF Frequency Band	60-64 GHz
	Tx/Rx Channels	3TX / 4RX (see Note 1)
	Modulation	FMCW
	Output Power per TX Channel (EIRP)	15 dBm
Connectivity and Integration	Cloud Protocols	MQTT, HTTP, HTTPS
	Wi-Fi	Wi-Fi 802.11 b/g/n (2.4 GHz)
		Station / SoftAP / Station + SoftAP
		Up to 150 Mbps (theoretical; actual performance depends on the network environment)
	Local Communication	UART (data format is defined by firmware and can support binary or JSON)
	Cellular Network (Optional)	LTE Cat.1 bis (4G, full-network support)
		Up to 10 Mbps downlink / up to 5 Mbps uplink (see Note 2)

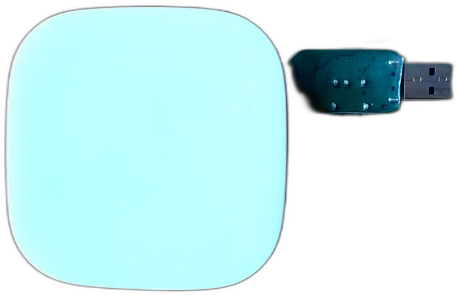
Category	Item	Specification
Hardware Architecture	Processing Architecture	Dual-chip heterogeneous architecture (mmWave radar SoC + main MCU)
	Radar Processing Unit	ARM Cortex-R4F + C674x DSP + Hardware Accelerator (HWA)
	Main Controller	ESP32-S3 (dual-core Xtensa LX7, up to 240 MHz)
	On-chip Memory	512 KB (main MCU) + 1.75 MB (radar SoC)
	PSRAM	8 MB PSRAM (connected to the main MCU)
	Flash Storage	8 MB (main MCU) + 4 MB (radar SoC, optional)
	I/O and Indicators	1x RGB LED, 1x key, 1x LED (optional)
	IMU (Optional)	Optional 6-axis gyroscope + 3-axis accelerometer
	Ambient Light Sensor (Optional)	Optional support
	Voice (Optional)	Optional support (1x speaker, 1x mic)

Note 1: For the “Tx/Rx Channels” parameter, up to 2TX simultaneous transmission is supported in 1.3 V mode; 3TX simultaneous transmission requires the 1V LDO bypass operating mode.

Note 2: The value is a theoretical peak. Actual performance depends on network coverage, signal quality, and carrier configuration.

4. Development and Flashing Orientation

To ensure successful flashing and serial communication, pay attention to the required `Type-C` orientation. When connecting the flashing debugger to a `Pro` device, align the `B` side of the debugger with the front side of the enclosure.

Platform	Alignment Diagram
Pro Device	

5. Interface Description

The PRO module interface description includes the flashing debugger Type-C, the status RGB LED, and the key input.

5.1 Type-C Interface on the Flashing Debugger

The USB-to-UART V1.3 debugger board is used for firmware flashing and serial console access. Both debugger variants use the same Type-C pin definition and communication capability.

If the port is used only for power delivery, the module does not distinguish between side A and side B. When used for communication, the module Type-C port is side-sensitive. The pin definition is as follows.

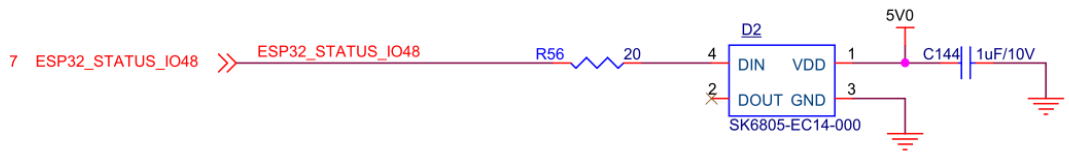
Type-C	Definition
A5	UART_RX
A6	RTS
A7	DTR
B8	UART_TX
A1/A12/B1/B12	GND
A4/A9/B4/B9	5V input

The USB-to-UART V1.3 flashing debugger pin definition is as follows.

Pin	Color	Signal
A5	Orange	RX
A6	Green	RTS
A7	Blue	DTR
B8	Yellow	TX
GND	Black	GND
VBUS	Red	5V

5.2 Status RGB LED Interface Reference

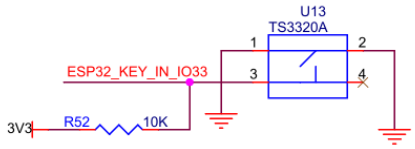
The PRO board provides an RGB status LED, and the main controller side uses ESP32-S3. The current reference diagram is shown below.



PRO status RGB LED interface reference

5.3 Key Interface Reference

The PR0 board provides a key input interface. The current reference diagram is shown below.



PRO key interface reference

6. Related Documents

1 RPX Series Usage Guide